

Input window

$\tilde{\mathbf{f}}_l, \tilde{\omega}_l$ [t_a - 2 s, t_b + 2 s]
200 Hz grey: a still phase

Anchor-frame pre-integration

$$\mathbf{R}_{a \leftarrow k} = \mathbf{R}_{CI} \mathbf{Q}_a^\top \mathbf{Q}_k \mathbf{R}_{CI}^\top, \quad \mathbf{f}_k^{C_a} = \mathbf{R}_{a \leftarrow k} \mathbf{R}_{CI} \tilde{\mathbf{f}}_{l,k}$$

$$\hat{\boldsymbol{\mu}}_f = \langle \mathbf{f}_k^{C_a} \rangle_{\text{window}}, \quad \ell_k = \mathbf{f}_k^{C_a} - \hat{\boldsymbol{\mu}}_f$$

$$\Delta \mathbf{v}_k = \Sigma \ell, \quad \Delta \mathbf{d}_k = \Sigma \Sigma \ell$$

no learned parameters, 24 channels

Shared trunk

Conv 7, stride 2

5 × dilated residual, d = 1, ..., 16

200 to 50 Hz, 0.42 M parameters

Conv 5, stride 2

BiGRU, 96 units

Three heads

Velocity $\hat{\mathbf{v}}_k$, upsampled to 200 Hz

Stillness $\sigma(s_k)$, BCE on reference speed

Log-variance $\eta_k \rightarrow \sigma_k$, Gaussian NLL

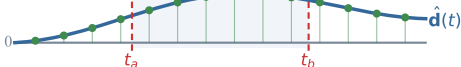
Structural kinematics (no free parameters except λ)

$$\hat{\mathbf{v}}_k \leftarrow \hat{\mathbf{v}}_k [1_k - \sigma(s_k)] \quad (\text{stillness-conditioned gate})$$

$$\hat{\mathbf{d}}_k = \frac{1}{f_s} \sum_{i=k} \hat{\mathbf{v}}_i + (\mathbf{R}_{a \leftarrow k} - \mathbf{I}) \lambda, \quad \hat{\mathbf{d}}(t_a) \equiv \mathbf{0}$$

$$u(\hat{\mathbf{d}}) \stackrel{a}{=} \kappa \sqrt{\sigma_k^2 + s_{\text{en}}^2} \quad \kappa \text{ fitted on validation}$$

Output: one displacement stream per window, supervised at every reference frame inside it



green: reference displacement at each usable chessboard frame, about 145 per window